

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VII

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Class :SYIT	Batch : 00
Date of Assignment :05/08/2026	Date/Time of Submission :05/08/2026, 9:30am

Hash Table:

Write a program to:

- a. Implement a hash table with separate chaining for collision handling.

Code:

```
size=5
table=[[] for _ in range(size)]
def insert(key,value):
    table[(key%size)].append((key,value))
insert(10,"Apple")
insert(20,"Poco")
insert(30,"Infinix")
insert(40,"Samsung")
insert(50,"Oppo")

print(table)
```

Output:

```
[[ (10, 'Apple'), (20, 'Poco'), (30, 'Infinix'), (40, 'Samsung'), (50, 'Oppo') ], [], [], [], [] ]
```

- b. Store and retrieve data from the hash table.

Code:

```

size=5
table=[[] for _ in range(size)]
def insert(key,value):
    table[(key%size)].append((key,value))
def search(key):
    for k,v in table[key%size]:
        if k==key:
            return v
    return "Not found"

insert(10,"Apple")
insert(20,"Poco")
insert(30,"Infinix")
insert(40,"Samsung")
insert(50,"Oppo")

print(table)

key=int(input("Enter the key to access element in hash table : "))
print(search(key))

```

Output:

```

===== RESTART: C:/Users/IT-03/dfb.py =====
[[ (10, 'Apple'), (20, 'Poco'), (30, 'Infinix'), (40, 'Samsun
'), (50, 'Oppo')], [], [], [], []]
Enter the key to access element in hash table : 50
Oppo

```

Curiosity Questions:

1. In a BST, which side (left or right) has **smaller numbers** than the root?
2. If we insert numbers 10, 20, 5 into a BST, which number will be at the root?
3. What will the inorder traversal of a BST give — sorted numbers or random numbers?
4. Where will the **largest number** in a BST always be found?
5. If the root of a BST is 50 and we search for 70, will we go left or right? Why?
6. If a BST has only **one node**, what is its inorder traversal?
7. Can we create a BST from characters like A, B, C instead of numbers?
8. If we insert numbers 5, 10, 15, 20 in order, will the tree be balanced or like a straight line?
9. In a BST, do we check the left child first or the right child first in inorder traversal?
10. If we delete the root of a BST, will the tree disappear?